MTH219 2022 S1 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

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* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH219

Timed Online Assignment – January Semester 2022

Fundamentals of Statistics and Probability

Thursday, 19 May 2022 **1:00 pm – 3:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

A survey of 20 resident visitors on the average frequency of their visits per quarter to the Gardens yield the sample:

1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, 4, 5, 4, 5, 6

It was discovered that one of the surveyed results was misplaced.

(a) Determine the following quantities for the sample of 20 frequencies. Comment on any assumption(s) used, and show detailed working,

(i) mode;

(ii) median;

(iii) mean;

(iv) inter-quartile range; and

(v) standard deviation.

(10 marks)

(b) It was later recovered that the missing frequency was 59. Comment on how the estimates in Question 1(a) would be affected when this recovered frequency is included in the calculation. Calculate the updated quantities.

(7 marks)

(c) Comment on which measure(s) would now be more appropriate when this new recorded frequency is used, and why?

(3 marks)

Question 2

Jenny has three bags, one white, one yellow, one orange. Each bag contains 20 identically sized balls. The white bag has 5 blue balls, the yellow bag has 10 blue balls, and the orange bag has 15 blue balls. The rest of the balls are red.

She now draws balls from the bags, one ball each time and replacing each ball picked before picking the next.

If a blue ball is picked from the white bag, Jenny next picks from the yellow bag, otherwise she next picks from orange bag. If a blue ball is picked from the yellow bag, Jenny next picks from the orange bag, otherwise she next picks from white bag. If a blue ball is picked from the orange bag, Jenny next picks from the white bag, otherwise she next picks from yellow bag.

If Jenny starts her draw from the white bag, compute the probability that

(a) the first 4 balls she drew are blue;

(4 marks)

(b) after 5 draws, she has not drawn from the orange bag.

(6 marks)

Question 3

A fair die is tossed twice and let X_1 and X_2 denote the scores obtained for the two tosses, respectively.

(a) Calculate $E[X_1]$ and show that $var(X_1) = \frac{35}{12}$.

(5 marks)

(b) Determine and tabulate the probability distribution of $Y = |X_1 - X_2|$ and show that $E[Y] = \frac{35}{18}$.

(7 marks)

(c) The random variable Z is defined by $Z = X_1 - X_2$. Comment with reasons (quantities concerned need not be evaluated) if each of the following statements is true or false.

(i) $E(Z^2) = E(Y^2)$

(ii) $var(Z) = var(Y)$

(8 marks)

Question 4

Let X be a binomial random variable with mean 4 and variance $\frac{4}{3}$.

(a) Apply the given information and show that the largest value that X can take is 6. Hence determine $P[X = 5]$.

(10 marks)

(b) Suppose X represents the number of eggs laid each year by a certain species of bird, and the probability that any egg laid will hatch is $\frac{3}{4}$. Calculate the probability of 5 or more eggs hatching in a single year from a random selected bird.

(10 marks)

Question 5

An advertising display contains a large number of light bulbs which are continually being switched on and off. Individual lights fail at random times, and each day the display is inspected, and any failed lights are replaced. The number of lights that fail in any one day period has a Poisson distribution with mean 2.2.

(a) What is the probability that no light will need to be replaced on a particular day?

(4 marks)

(b) What is the probability that at least four lights will need to be replaced over a stretch of 2 days?

(5 marks)

(c) What is the least number of consecutive days after which the probability of at least one light having to be replaced exceeds 0.9999?

(6 marks)

Question 6

(a) Darren scored a mark of 57 on the Miller Analogies Test. This test had a mean of 50 and a standard deviation of 5. Jennifer scored 120 on the WISC Intelligence Test. This test had a mean of 100 and a standard deviation of 15. Comparing their scores, comment on who had a better score? Explain your answer.

(6 marks)

(b) The performance scores (each score is an x-value) of three drivers were converted to standard scores. Comment on what each of the standard z-score indicate and determine the related implication.

(i) $z = 0.03$.

(ii) $z = 4.2$.

(iii) $z = -0.49$.

(9 marks)

----- END OF PAPER -----

1(a) mean of 19 scores = 3.052632

$\Rightarrow$ assume missing score is 3

$$x = (1,1,1,1,\ 2,2,2,2,\ 3,3,3,3,\ 4,4,4,4\ 5,5,5,6)$$

$$\text{Mode} = \{1,2,3\}$$

$$\text{mean} = \frac{1}{n}\sum_{i=1}^{n} x_i = 3.05$$

$$\begin{aligned}\text{Median} &= x_{0.5(21)} \\ &= x_{10.5} \\ &= x_{10} + 0.5(x_{11} - x_{10}) \\ &= 3\end{aligned}$$

$$\begin{aligned}\text{inter-quartile range} &= x_{0.75(21)} - x_{0.25(21)} \\ &= x_{5.75} - x_{5.25} \\ &= (x_{15} + 0.75(x_{16} - x_{15})) - (x_5 + 0.25(x_6 - x_5)) \\ &= 4 - 2 \\ &= 2\end{aligned}$$

$$\begin{aligned}\text{sample variance} &= \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \\ &= 2.36576\end{aligned}$$

$$\begin{aligned}\text{standard deviation} &= \sqrt{\text{sample variance}} \\ &= 1.538112\end{aligned}$$

1(b) $x = (1,1,1,1,$
$2,2,2,2,$
$3,3,3,$
$4,4,4,4$
$5,5,5,$
$59)$

mode $= \{1,2,4\}$

mean $= \frac{1}{n}\sum_{i=1}^{n} x_i = 5.8$

interquartile range $= x_{15.75} - x_{5.25}$

$= [x_{15} + 0.75(x_{16} - x_{15})] - [x_5 + 0.25(x_6 - x_5)]$

$= 4.75 - 2$

$= 2.75$

sample variance $= \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 158.8711 \simeq 158.9$

standard deviation $= \sqrt{\text{sample variance}}$

$= 2.160247$

$\simeq 2.16$

2.

bag	blue ball	red ball
white	5	15
yellow	10	10
orange	15	5

(a) P(first 4 balls are blue)

$= P(\text{white} \to \text{yellow}) \times P(\text{yellow} \to \text{orange}) \times$
$\cdot P(\text{orange} \to \text{white}) \times P(\text{white} \to \text{yellow})$

$= \frac{5}{20} \times \frac{10}{20} \times \frac{15}{20} \times \frac{5}{20}$

$= \frac{1}{4} \times \frac{1}{2} \times \frac{3}{4} \times \frac{1}{4}$

$= \frac{3}{128}$

2(b) If she has not drawn from orange box, then she must be alternating between white and yellow bags for 5 draws.

P(After 5 draws, she has not drawn from orange bag)
= P(white → yellow) × P(yellow → white) × P(white → yellow) × P(yellow → white)

$= P^2(\text{white} \to \text{yellow}) \times P^2(\text{yellow} \to \text{white})$

$= \left[\frac{5}{20}\right]^2 \times \left[\frac{10}{20}\right]^2$

$= \left[\frac{1}{4}\right]^2 \times \left[\frac{1}{2}\right]^2$

$= \frac{1}{2^6}$

3(g) $E(X_1) = \frac{1}{n}\sum_{i=1}^{n} i$ where $n=6$

$$= \frac{1}{n} \cdot \frac{n(n+1)}{2}$$

$$= \frac{n+1}{2}$$

$$= \frac{7}{2}$$

$E(X_1^2) = \frac{1}{n}\sum_{i=1}^{1} i^2$ where $n=6$

$$= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{(n+1)(2n+1)}{6}$$

$$= \frac{7 \times 13}{6}$$

$$= \frac{91}{6}$$

$$Var(X_1) = E(X_1^2) - [E(X)]^2$$

$$= \frac{91}{6} - \left(\frac{7}{2}\right)^2$$

$$= \frac{182}{12} - \frac{147}{12}$$

$$= \frac{35}{12}$$

3(b) $Y = |X_1 - X_2|$

$$P(Y=0) = P(X_1 = X_2)$$
$$= \sum_{i=1}^{6} P(X_1 = i) \cdot P(X_2 = i)$$
$$= \sum_{i=1}^{6} \left(\frac{1}{36}\right) = \frac{1}{6}$$

$$P(Y=1) = 2\,P(X_2 = X_1 + 1)$$
$$= 2 \sum_{i=1}^{5} P(X_1 = i) \cdot P(X_2 = i+1)$$
$$= 2 \sum_{i=1}^{5} \left(\frac{1}{36}\right)$$
$$= \frac{5}{18}$$

$$P(Y=2) = 2\,P(X_2 = X_1 + 2)$$
$$= 2 \sum_{i=1}^{4} P(X_1 = i) \cdot P(X_2 = i+2)$$
$$= 2 \sum_{i=1}^{4} \left(\frac{1}{36}\right).$$
$$= \frac{2}{9}$$

$$P(Y=3) = 2\,P(X_2 = X_1 + 3)$$
$$= 2 \sum_{i=1}^{3} P(X_1 = 1) \cdot P(X_2 = i+3)$$
$$= 2 \sum_{i=1}^{3} \frac{1}{36}$$
$$= \frac{1}{6}$$

3(b) $P(Y=4) = 2\,P(X_2 = X_1 + 4)$

$$= 2 \sum_{i=1}^{2} P(X_1 = i) \cdot P(X_2 = i+4)$$

$$= 2 \sum_{i=1}^{2} \left(\frac{1}{36}\right)$$

$$= \frac{1}{9}$$

$P(Y=5) = 2\,P(X_2 = X_1 + 5)$

$$= 2 \sum_{i=1}^{1} P(X_1 = i) \cdot P(X_2 = i+5)$$

$$= 2 \sum_{i=1}^{1} \left(\frac{1}{36}\right)$$

$$= \frac{1}{18}$$

k	0	1	2	3	4	5
$P(Y=k)$	1/6	5/18	2/9	1/6	1/9	1/18

$$E(Y) = \sum_{k=0}^{5} k \cdot P(Y=k)$$

$$= \left[\frac{5}{18} + \frac{4}{9} + \frac{3}{6} + \frac{4}{9} + \frac{5}{18}\right]$$

$$= \frac{35}{18}$$

3 (c) i. $Y = |X_1 - X_2|$ $\quad Z = X_1 \cdot X_2$

$$E(Z^2) = E(|Z|^2)$$
$$= E(|X_1 - X_2|^2)$$
$$= E(Y^2)$$

The statement is True

ii $var(Z) = var(X_1 - X_2)$
$$= var(X_1) + var(X_2)$$
$$= 2var(X)$$

$$var(Y) = E(Y^2) - [E(Y)]^2$$
$$= E(|X_1 - X_2|^2) - [E(|X_1 - X_2|)]^2$$
$$= E((X_1 - X_2)^2) - 0$$
$$= E(X_1^2 - 2X_1X_2 + X_2^2)$$
$$= E(X_1^2) - 2E(X_1)E(X_2) + E(X_2^2)$$
$$= E(X^2) - 2E^2(X) + E(X^2)$$
$$= 2E(X^2) - 2E^2(X)$$
$$= 2[E(X^2) - E^2(X)]$$
$$= 2\,var(X)$$

$\therefore var(Y) = var(Z)$

The statement is true

4(a) $X \sim Bin(n,p)$

$np = 4$, $np(1-p) = \frac{4}{3}$

$\Rightarrow 1-p = \frac{4}{3np} = \frac{4}{3(4)} = \frac{1}{3}$

$p = \frac{2}{3}$

$n = \frac{4}{p} = \frac{3}{2} \times 4 = 6$

$\therefore$ max $X = n = 6$

$\therefore P(X=5) = \binom{6}{5} p^5 (1-p)$

$= \text{dbinom}(5,6,p)$

$= 0.2633745$

(b) $X \sim Bin(6, \frac{2}{3})$

let Y be no. of eggs that hatch

$Y \sim Bin(X, \frac{3}{4})$

P(5 or more eggs hatching in a single year by a randomly selected bird)

$= P(X=5, Y=5) + P(X=6, Y \geq 5)$

$= P(X=5) . P(Y=5 \cap X=5) + P(X=6) . P(Y \geq 5 \cap X=6)$

$= P(X=5) . P(Y=5 \cap X=5) + P(X=6) . [1 - P(Y \leq 4 \cap X=6)]$

$= \text{dbinom}(5,6,\frac{2}{3}) \times \text{dbinom}(5,5,\frac{3}{4}) +$
$\text{dbinom}(6,6,\frac{2}{3}) \times (1 - \text{pbinom}(4,6,\frac{3}{4}))$

$= 0.109375$

$\simeq 0.109$

5. let X_T be no. of lights that fail in T-day period

$X_T \sim Po(2.2T)$

(a) $X_1 \sim Po(2.2)$

P(no light will be replaced on a particular day)

$= P(X_1 = 0)$

$=$ dpois(0, 2.2)

$= 0.1106032$

$\simeq 0.111$

(b) $X_2 \sim Po(4.4)$

P(at least 4 lights will need to be replaced in a stretch of 2 days)

$= P(X_2 \geq 4)$

$= 1 - P(X_2 \leq 3)$

$= 1 -$ ppois(3, 4.4)

$= 0.6405522$

$\simeq 0.641$

(c) $X_T \sim Po(2.2T)$

$P(X_T \geq 1) > 0.9999$

$1 - P(X_T = 0) > 0.9999$

$P(X_T = 0) < 0.0001$

$$\frac{e^{-2.2T}(2.2T)^0}{0!} < 0.0001$$

$e^{-2.2T} < 0.0001$

$e^{2.2T} > 10000$

$\ln e^{2.2T} > \ln 10000$

$2.2T > \ln 10000$

$T > \frac{\ln 10000}{2.2}$

$T > 4.186518$

$\therefore$ At least 5 consecutive days

6. (a) let X be score for Miller Analogies Test
let Y be score for WISC Intelligence Test

$X \sim N(50, 5^2)$ $\quad$ $Y \sim N(100, 15^2)$

Percentile for Darren's score
= pnorm(57, 50, 5)
= 91.9%

Percentile for Jennifer's score
= pnorm(120, 100, 15)
= 90.9%

Darren has the better score because his percentile is higher than Jennifer's

(b) i. $z = 0.03$
percentile $= P(Z \leq 0.03)$
= pnorm(0.03, 0, 1)
= 51.2%

ii. $z = 4.2$
percentile $= P(Z \leq 4.2)$
= pnorm(4.2, 0, 1)
= 99.99867%

iii. $z = -0.49$
percentile $= P(Z < -0.49)$
= pnorm(-0.49, 0, 1)
= 31.20669%

The drivers with $z = 0.03, 4.2$ passed the exam.
The driver with $z = -0.49$ failed the exam
The driver with $z = 4.2$ is the top scorer.